Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology University
Avadi, Chennai

NAAC A++

CATEGORY 1 DEEMED TO BE UNIVERSITY by the UGC

SCHOOL OF COMPUTING

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT

Name	: S.chavan Teja
VTU Number	: 28581
Course Code/ Course Name	: 10212CS211/Artificial Intelligence Techniques
Year	: Summer 2026-27
Course Handling Faculty	: Dr.S.Pragadeeswaran TTS3903
Slot	: S6L1
Course Coordinator	: Dr. R. RoselinKiruba TTS3227
Assignment Group Number	: 4
Date	: 26/10/26

Assignment Mark Allocation

Criteria	Total Mark	Mark Achieved
Concept Understanding (A)	1	1
Accuracy & Correctness (B)	1	1
Presentation Clarity & Communication (C)	1	1
Relevance & Completion (D)	1	1
Timeliness (E)	1	1
Total (A+B+C+D+E)	5	5

General Comments:

Write all mathematical expressions and symbols clearly

Faculty Signature

Faculty Name & TTS:

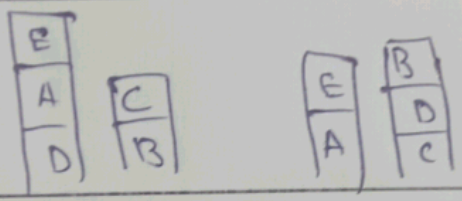
Dr.S.Pragadeeswaran

TTS:3903

CO NOs	Course Outcomes	domain(Based on revised Blooms taxonomy)
CO1	Identify the problem-solving techniques by using State Space search and heuristic search approach	K3
CO2	Use evolutionary and advanced search techniques to provide optimized solution.	K3
CO3	Solve unified planning approach and Constraint-satisfaction problems.	K3
CO4	Apply Logical knowledge representation for rules and fact-based approach.	K3
CO5	Build the logic of Uncertainty and Reasoning using resolution methods.	K3

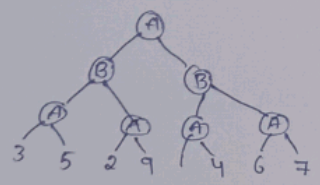
Question Group:

GROUP -1

S.No	List of Questions	CO	K Level
1	Apply the Minimax algorithm to determine the Optimal move for player A (the root)	CO1	K2
2	Slove the following Blocks world problem using goal stack planning show each step to the solution.	CO2	K3
3		CO3	K2
4	Express the following 4 most specific concepts that the individuals Schin and sucheta belong to?	CO4	K2
5	Formulate the given tic tac toe problem using heuristic function.	CO5	K3

y the Minimax algorithm to determine the optimal move for figure A(the root)

-A)



Step1:- Evaluate terminal nodes(leaf values)

Leaf values are already given: 3,5,2,9,1,4,6,7

Step2:- Propagate values using Minimax (bottom-up)

- Level2(A nodes):
  - Left A(min) = min(3,5) = 3
  - Right A(min) = min(2,9) = 2
  - Left A(min) = min(1,4) = 1
  - Right A(min) = min(6,7) = 6

- Level1(B nodes-max):
  - Left B = max(3,2) = 3
  - Right B = max(1,6) = 6

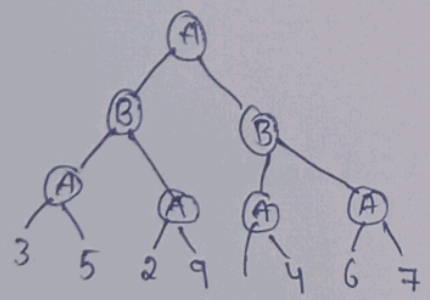
Step3:- Root A(min):

A chooses the minimum of (3,6) = 3

Optimal move for player A = left subtree

Value = 3

Ply the Minimax algorithm to determine the optimal move for figure A(the root)



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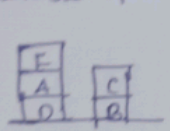
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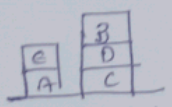
the following blocks world problem using Goal stack planning.

Problem: show each step to the solution. Show every step in details.

a) Available operators are: stack(x,y) unstack(x,y) pickup(x) putdown(x).



Initial state (start state)



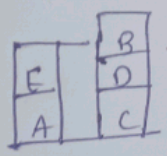
Goal State

Goal Stack (top → bottom): B, D, C, E, A

Steps:

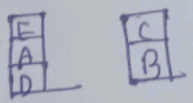
1. unstack(C,B) → C is clear, B is clear (C on table)
2. stack(C, D) → D is clear, (C on D)
3. Unstack(E,A) → E is clear, A is clear (E on table)
4. stack (E,B) → B is clear, (E on B)
5. stack (A,D) → D is clear, (A on D)

Final state:-

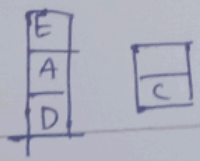


Step by - Step States:-

Initial:



After 1: unstack(C, B)



The ADC University employs Sucheta.

Formal: employs (ADC, Sucheta)

d) Sucheta has at least two cousins and all her cousins are fans only of cricketers and are all fans to sachin.

Formal: ≥ 2 cousin(sucheta, x) $\wedge \forall x$ (cousin(sucheta, x) $\rightarrow$ (Fan(x, cricketer) $\wedge$ Fan(x, sachin))

e) ABC university is a university that employs only Indian citizens who have phd degree.

Formal: University(ABC) $\wedge \forall x$ (employs(ABC, x) $\rightarrow$ (Indian citizen(x) $\wedge$ phD(x))

Most specific concepts: person, fan, cricketer, Indian citizen, phD

Consider the following axioms:

1. All hounds howl at night.
2. Anyone who has any cats will not have any mice.
3. Light sleepers do not have anything which howls at night.
4. John has either a cat or a hound
5. (Conclusion) If John is a light sleeper, then John does not have any mice.

Resolution method:-

1. From 4: cat(John) $\vee$ Hound(John)
2. From 1: Hound(x) $\rightarrow$ HowlsAtNight(x)

Instantiate (x = John) $\Rightarrow$ Hound(John) $\rightarrow$ HowlsAtNight(John)

From 3: LightSleeper(x) → ¬HowlsAtNight(x)

Instantiate x = John ⇒ LightSleeper(John) → ¬HowlsAtNight(John)

4. From 2 & 3: Cat(John) → ¬Mice(John)

LightSleeper(John)

5. Assume LightSleeper(John)

From 4: ¬Mice(John)

conclusion:- If John is light sleeper, then John does not have any mice.

Resolution refutation (alternative):-

Negate conclusion: LightSleeper(John) ∧ Mice(John)

Add to KB and derive contradiction:

1. LightSleeper(John) → ¬HowlsAtNight(John)
2. Hound(John) → HowlsAtNight(John) (from 4)
3. ¬Mice(John) → Cat(John) (from 2)
4. Cat(John) ∨ Hound(John) (from 4)
5. Assume LightSleeper(John) and Mice(John)

→ contradiction (requires both cat and hound)

∴ conclusion is entailed.

From 3: LightSleeper(x) → ¬HowlsAtNight(x)

Instantiate x = John ⇒ LightSleeper(John) → ¬HowlsAtNight(John)

4. From 2 & 3: Cat(John) → ¬Mice(John)

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5. Assume LightSleeper(John)

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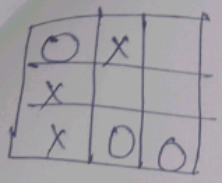
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5. Assume LightSleeper(John) and Mice(John)

→ contradiction (requires both cat and hound)

∴ conclusion is entailed.

the given tic tac toe problem using heuristic function.



Heuristic Function:-

h(n) = (number of open lines for X) - (number of open lines for O)

(Higher value is better for X).

open lines for X = 4 (rows: 1, 2, 3; columns: 2, 3; diagonal: 2)

open lines for O = 2 (columns: 1, 2)

h(n) = 4 - 2 = 2

So, the heuristic value of this state is 2 (favorable for X).